

George Gu

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Education

University of Michigan

May 2028

Bachelor of Science in Engineering, Computer Science

Ann Arbor, MI

- **GPA:** 3.81/4.00
- **Coursework:** Machine Learning, Artificial Intelligence, Reinforcement Learning, Operating Systems, Theory of Computation, Computer Organization, Web Systems, Computer Security, Data Structures & Algorithms

Experience

Capital One

June – August 2026

Software Engineer Intern

Richmond, VA

- Delivered an end-to-end **production** asset certification feature: replacing manual outreach across **2000+** assets with a secure **PUT** endpoint under three-tier authorization and matching React components.
- Built **2 EventBridge**-scheduled Lambdas, automating certification reminder emails and ServiceNow CMDB sync daily, eliminating up to **8 months** of data staleness from a prior 1-time load.
- Authored idempotent **Flyway** PostgreSQL migrations adding a CMDB caching layer and tuning SQL for compliance reporting.
- Configured **CI/CD** across dev, QA, and prod using Jenkins, IAM roles and CloudFormation stacks.

BoilerVault Storage LLC

January – June 2026

Founding Engineer (Contract)

West Lafayette, IN

- Built a multi-tenant **FastAPI**/PostgreSQL/Next.js operational platform for a **3-campus** storage business, implementing JWT auth across 20+ REST endpoints validated by a **318-test** backend suite.
- Reconciled out-of-order payments via webhook pipelines across **4 Stripe accounts** and **3 WordPress sites** over **\$130K+** in revenue.
- Automated unmatched charge reconciliation saving **3+** hours of manual work per week by storing and sweeping oldest-first.
- Migrated 475 legacy bookings into **2,000+** records across 5 tables, replacing a failing Zapier Google Sheets workflow.

Nexteer Automotive

May – August 2025

Software Engineer Intern

Saginaw, MI

- Built an **LLM-powered** IDE extension in Python and TypeScript parsing **300+** internal engineering guidelines, automating compliance checks across C/H files via Azure AI and GitHub Copilot APIs.
- Tuned prompt pipelines and few-shot strategies to **95% violation-detection accuracy** on embedded steering code.
- Deployed to **26 sites**, saving engineers an estimated **8 hours** of manual code review per week.

Villanova University

June – September 2023

Data Engineer

Villanova, PA

- Engineered data pipelines in R, applying PCA and clustering techniques to analyze **28,000+** pathogen isolates across **10+ years**, uncovering essential trends published as first author in *Antibiotics*.

Projects

MeloChron

July – August 2026

Machine Learning Researcher

Ann Arbor, MI

- Architected a causal self-attention encoder summarizing track history and 2-layer MLP with **468K parameters** predicting whether a listener returns to a new track, evaluated on **100,000 listeners** via PR-AUC.
- Provisioned a cloud **RTX 4090** instance to train the **PyTorch** model, cutting a multi-day run to hours while scaling training data 7x.
- Iterated past a strong baseline (**0.4212**) to an audio-content model at **0.4820 PR-AUC**, passing a paired bootstrap significance test.

Scalable Search Engine

January – April 2026

Systems Developer

Ann Arbor, MI

- Built a multi-stage **MapReduce** pipeline indexing **3,000+ documents**, computing TF-IDF scores and per-document normalization factors across a parallel, multi-job architecture.
- Served **3 partitioned index segments** from a Flask API, ranking by PageRank-weighted cosine similarity and deployed to AWS.

Resume Screener

January – May 2025

Machine Learning Developer

Ann Arbor, MI

- Built an **NLP** pipeline over **1,000+** labeled resumes with tokenization and TF-IDF vectorization for fit scoring at **92%** accuracy.
- Ranked resumes with **scikit-learn/TensorFlow** models and **RAG** retrieval, deployed via Streamlit to **100+ users**.

Skills/Additional

Languages: Python, SQL, C/C++, Java, JavaScript, TypeScript, HTML, CSS

Frameworks/Developer Tools: PyTorch, TensorFlow, scikit-learn, Git, Docker, AWS, Unix, Node.js, React.js, Flask, PostgreSQL

Interests: Philosophy and Metaphysics, Game Design, Boulderling, Skateboarding, Golf, Guitar, Calisthenics

Publication

Gu G, Pei H, Zhou A, Fan B, Zhou H, Choi A, Huang Z, “A Comprehensive Study of Historical Detection Data for Pathogen Isolates from U. S. Cattle”, 2023, *Antibiotics*, 12, 1509.

doi: <https://doi.org/10.3390/antibiotics12101509>